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Mechanistic Considerations for Battery Charging Protocol Design

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ABSTRACT

The rapid growth of lithium-ion batteries (LIBs) applications drives the need for fast-charging solutions ensuring speed, safety, durability, and performance. Such charging protocol design needs to be guided by mechanistic understanding of degradation pathways, ionic transport limitations, and thermal constraints. However, in practice, many charging protocols used in commercial electronics and electric vehicles (EVs) have limited mechanistic transparency. In this review, we adopt a reverse perspective by extracting mechanistic insights from practical charging protocols to inform future design. To this end, standardized fast-charging protocols and those implemented in real-world applications such as smartphones and EVs are analyzed to examine how their voltage–current profiles evolve with state-of-charge (SOC) and to reflect distinct design rationales. These features are further examined in terms of SOC-dependent physical and chemical transformations in electrode materials, kinetic limitations such as polarization and reaction heterogeneity influenced by charging protocol design, and distinct heat generation patterns governed by protocol characteristics. Advanced characterization techniques are then highlighted for providing real-time insights into structural transitions, diffusion kinetics, and heat evolution during fast charging. Finally, future protocol design may be informed by multiscale material modelling, real-time sensing for adaptive control, and data-driven optimization to support the development of advanced fast-charging systems.

1 | Introduction

Lithium-ion batteries (LIBs) have become indispensable in modern society, powering electric vehicles (EVs), portable electronics, and stationary energy storage systems. As these technologies scale rapidly, the demand for fast-charging capabilities has grown in parallel, aiming to reduce downtime and improve user experience. However, fast charging imposes severe stresses on battery materials, often accelerating degradation processes such as lithium plating, structural fatigue, and thermal instability, ultimately compromising performance, lifespan, and safety [1–3].

To address these challenges, extensive research and development (R&D) have focused on designing fast-charging protocols that deliver high charging rates while preserving electrochemical and thermal stability [4–6]. Various innovative approaches, including pulse charging [7–9], multi-stage charging [10, 11], and variable-rate charging [12, 13], have been explored to accelerate

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charging and suppress lithium plating. A notable advancement in this area is the development of adaptive and smart charging systems, which dynamically adjust charging parameters based on real-time sensing data such as the battery's temperature, state of charge (SOC), and state of health (SOH) [14–17]. These approaches, often integrated with advanced battery management systems (BMS), enhance long-term performance by preventing overcharging, overheating, and degradation [18, 19]. Moreover, effective thermal management remains crucial, with protocols designed to limit thermal stress, prevent accelerated degradation, and ensure safe operation across a wide range of ambient temperatures [20, 21]. The complexity of LIBs charging protocols extends beyond individual cells to encompass device, pack, and system-level interactions. Factors such as material properties, cell design, and system packaging critically influence overall battery behaviour, underscoring the importance of understanding the relationship between microscopic properties and macroscopic performance [22, 23]. However, despite delivering reasonable performance, many protocols currently used in EVs and portable electronics offer very limited insight into their underlying mechanisms. This lack of mechanistic clarity hinders their generalization to new chemistries and architectures, making a protocol-informed mechanistic perspective essential for designing next-generation fast-charging strategies that are both efficient and universally applicable.

Based on analyzing the protocols deployed in real-life applications, this review provides a reverse perspective on fast charging. We emphasize the mechanistic interpretation of current–voltage features observed in both standardized and commercial protocols. Charging behaviors are categorized and analyzed in relation to SOC-dependent structural and compositional changes in electrode materials, kinetic limitations such as polarization and inhomogeneity, and heat generation characteristics shaped by protocol design. These mechanistic aspects are examined through a review of advanced characterization techniques that enable the analysis of structural evolution, electrochemical kinetics, and thermal responses under fast-charging, with particular emphasis on operando and in-situ methods. In addition, strategies such as multiscale modelling, real-time sensing, and data-driven analysis are briefly introduced to contextualize the future development of more intelligent and adaptive charging protocols. A comprehensive framework outlining these interrelated aspects is illustrated in Figure 1.

2 | Existing Battery Charging Protocols

The design of battery charging protocols primarily centers on the regulation of the current as functions of time during the charging process. This aspect of battery management offers a high degree of freedom, as there are numerous parameters that can be tuned to affect the outcomes. The careful adjustment of these parameters is crucial, as it can significantly influence the overall performance and safety of the battery. To provide a comprehensive understanding, several widely utilized charging methods are schematically illustrated in Figure 2a,b. They are categorized into two types: 1) charging protocols with predefined constant current stages (Figure 2a) and 2) charging protocols with dynamic current regulations based on the battery's instantaneous status (Figure 2b). As it will be discussed below, each of these methods clearly demonstrates its own set of advantages, featuring thoughtful considerations behind its design.

In the first category, **Constant Current-Constant Voltage (CC-CV)** charging method is the most widely used protocol for lithium-ion batteries. It involves two distinct phases: the constant current phase, where the battery is charged at a fixed current until it reaches a predetermined voltage threshold, followed by the constant voltage phase, during which the charging current gradually decreases while maintaining a constant voltage until the battery is fully charged. The simplicity and effectiveness of this method make it the popular choice for most lithium-ion battery applications. It provides a good balance between charging speed and battery longevity by reducing the current as the battery approaches full charge, thereby minimizing stress on the battery. However, the high current during the constant-current phase can lead to significant heat generation, which may accelerate battery degradation [29, 30]. Additionally, the constant-voltage phase can be lengthy, particularly as the charging current decreases, which limits the overall speed of the charging process. **Multistage Constant Current-Constant Voltage (MCC-CV)** method enhances the CC-CV approach by dividing the constant current charging phase into multiple sub-stages. Each stage applies a different constant current level before transitioning to the constant voltage phase. This staged approach allows for better control of the charging process, improving charging efficiency and reducing the risk of overheating. The gradual reduction of current helps reduce stress on battery materials, potentially extending the battery's cycle life [31]. While MCC-CV could potentially improve safety and battery life, the stepwise reduction in current can result in longer overall charging times compared to simpler methods. **Boost charging** is a notable development in this field [32–34]. It is designed to rapidly charge a battery by applying a very high current in the early stage of the charging process when the battery's SOC is low. This method significantly reduces the initial charging time by taking advantage of the battery's ability to accept higher currents at relatively low SOC levels. By quickly charging the battery to a substantial portion of its capacity, boost charging is particularly useful in scenarios where quick top-ups are needed. However, the high initial current can pose significant safety risks if not properly managed, including the potential for thermal runaway. If the transition from high to low current is not carefully controlled, it may lead to increased wear on the battery materials, reducing the overall lifespan of the battery. **Pulse charging** is another relatively novel approach, bringing in additional degrees of freedom. It involves applying intermittent pulses of current to the battery, rather than a continuous flow. This method introduces rest periods between pulses, which can help alleviate internal resistance buildup and reduce the risk of lithium plating. The rest periods allow for the redistribution of lithium ions within the electrode materials, which can mitigate some of the adverse effects associated with fast charging, such as electrochemical polarization and heterogeneity [35, 36]. Pulse charging can shorten overall charging time without compromising battery health. However, the effectiveness of pulse charging can vary significantly depending on the battery chemistry and the specific design of the pulse structure, such as frequency and amplitude [37]. Additionally, pulse charging can have a rather large and varying instantaneous heat generate rate compared to continuous charging methods, potentially accelerating battery aging. **Constant Power-Constant Voltage (CP-CV)** is very

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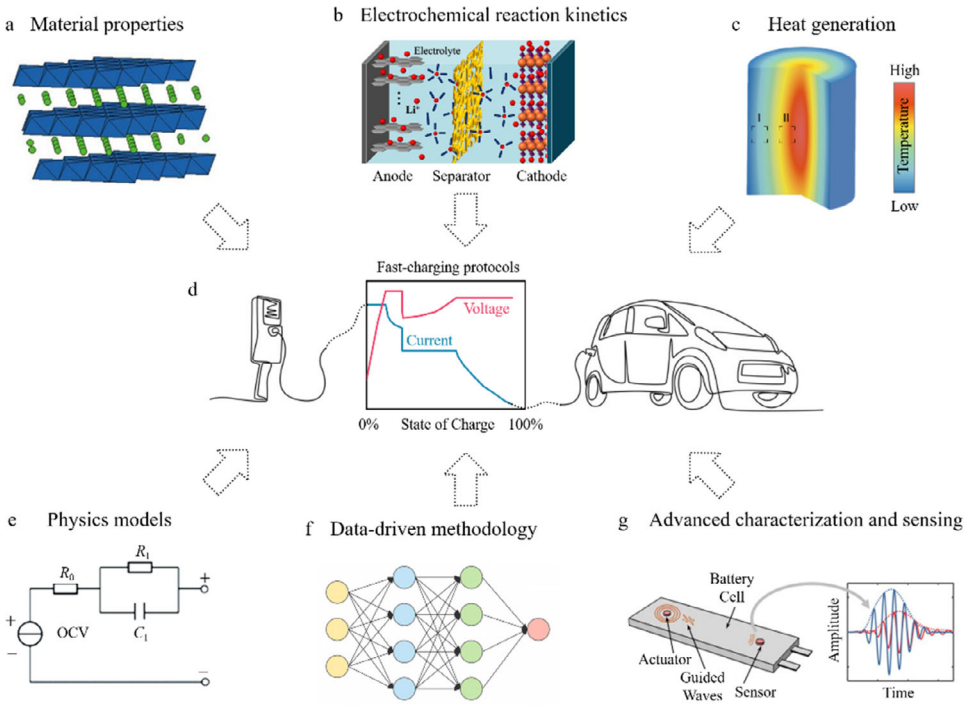


FIGURE 1 | Comprehensive framework for smart fast charging technologies [24–27]. a) Material properties. Reproduced with permission [24]. Copyright 2002, Nature Publishing Group. b) Reaction kinetics. Reproduced with permission [25]. Copyright 2024, MDPI. c) Heat generation. Reproduced with permission [26]. Copyright 2019, Nature Publishing Group. d) Charging protocols. e) Model-based strategies. f) Machine learning-based strategies. g) Characterization techniques. Reproduced with permission [27]. Copyright 2018, Elsevier.

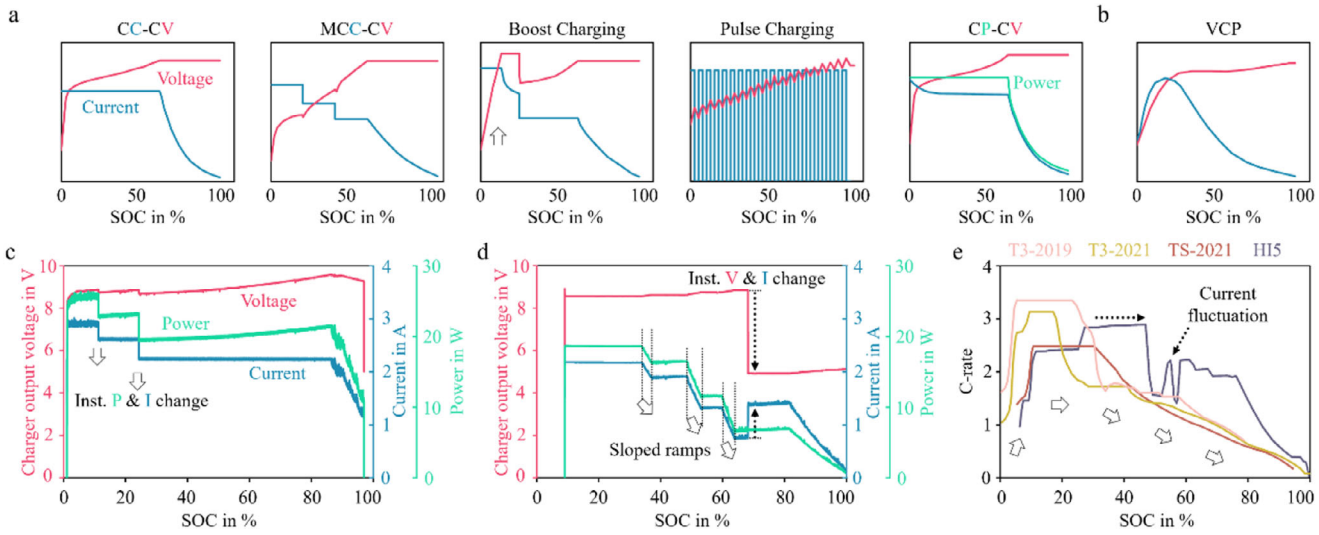


FIGURE 2 | Schematic illustrations of a few common LIBs charging protocols, including a) Constant Current-Constant Voltage (CC-CV), Multistage Constant Current-Constant Voltage (MCC-CV), Boost Charging, and Pulse Charging, and b) Constant Power-Constant Voltage (CP-CV) and Variable Current Profile (VCP). c) The charging protocol adopted by a Xiaomi Redmi Note 12 smartphone. d) The charging protocol adopted by an Apple iPhone 16 Pro smartphone. e) The charging protocols used by a few electric vehicles: Tesla Model 3 LR AWD 2019 (T3-2019), Tesla Model 3 LR AWD 2021 (T3-2021), Tesla Model S Plaid 2021 (TS-2021), and Hyundai Ioniq 5 (HI5). Replotted from data reported by InsideEVs [28]. Accessed October 2025.

similar to CC-CV except for that the first phase is executed with a constant power instead of constant current [38]. The constant power supply results in a gradual decrease in charging current as the voltage increases with the SOC increase. CP-CV could also be advantageous from the charging infrastructure perspective, which sets another practical limitation on the charging speed.

In the second category, **Variable Current Profile (VCP)** charging dynamically adjusts the charging current based on real-time monitoring of the battery's state, such as temperature, SOC, and internal resistance [39–41]. This method tailors the charging current to the battery's specific needs at each stage of the charging process, maximizing efficiency while minimizing risks associated

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with fast charging. VCP allows for continuous optimization of the charging process, ensuring that the battery receives the highest possible current that is safe at any given time. By tailoring the current to the battery's real-time condition, VCP can reduce the risks of overheating, lithium plating, and other forms of degradation, thus extending the battery's life. However, implementing VCP requires advanced BMS capabilities and real-time battery sensing, making it more complex and potentially more costly than simpler charging methods. In addition, the control algorithm can be sophisticated, and its optimization is not straightforward. The effectiveness of VCP is heavily dependent on the accuracy of the sensors used to monitor the battery's state, which may require careful calibration and maintenance.

The charging methods outlined earlier each have specific strengths and limitations, making them suitable for various applications depending on the requirements of the device or vehicle being powered. In real-world scenarios, however, charging protocols tend to be more intricate. Here, we analyze a few practical examples, including two commercial smartphones (Figure 2,d) and several electric vehicles (Figure 2e). In our experiments, we monitored the current, voltage, and charging power profiles of two arbitrarily selected commercial smartphones, the Xiaomi Redmi Note 12 and the Apple iPhone 16 Pro, during their respective charging processes. The data reveals that both devices incorporate multiple CC stages, followed by a gradual current ramp-down as the cells approaches the full SOC. Despite these similarities, there are key differences between the two cases. First, the transitions between consecutive CC stages in Figure 2c occur instantaneously, whereas those in Figure 2d feature a gradual ramp. Second, the externally applied charging voltage (~9 V) represents the adapter output, which is substantially higher than the cell-level voltage (3.7–4.2 V) regulated by the internal power-management circuitry during the initial charging stage. A distinct voltage drop is observed when the SOC reaches predefined thresholds. In Figure 2c, this voltage change occurs near the end of the charging process, while in Figure 2d, it takes place at around 70% SOC. Notably, this voltage drop in Figure 2d is accompanied by an instantaneous increase in current, as the charging power remains constant between approximately 65% and 80% SOC. Unlike large-format battery systems, smartphone batteries exhibit no visible current ramp-up at low SOC due to their small capacity, low operating voltage, and rapid electronic regulation. The power-management integrated circuit (PMIC) stabilizes voltage and current within milliseconds through soft-start control, effectively bypassing the need for a gradual electrochemical activation stage.

In contrast, the complexity increases significantly in larger systems such as EVs. The much higher energy capacity and voltage of EV battery packs necessitate a controlled ramp-up phase at the onset of charging to suppress abrupt overpotentials, balance inter-module voltages, and allow the BMS to verify cell conditions before high-current operation. At the material level, this behavior is further associated with the low lithium-ion diffusivity and sluggish interfacial kinetics of electrode materials near full discharge, a topic that will be elaborated on in the next section. As shown in Figure 2e, these electrochemical and system-level factors collectively give rise to an initial current ramp-up segment, followed by sustained high-C-rate charging (approximately 2C–3C) until the battery pack reaches about 30%–40% SOC, and then a gradual current taper as SOC increases. An exception to this trend is the Hyundai Ioniq 5 (HI5), which extends the high C-rate charging phase to an SOC of around 50%. Reflecting this extended high-power window, the HI5 protocol exhibits an average C-rate of approximately 2.2 between 20%–80% SOC, which is substantially higher than the 1.3 to 1.5 C observed for Tesla models. While high-current charging enables rapid energy storage, it also intensifies polarization, heat generation, and reaction heterogeneity within the cell. To mitigate these effects, the HI5 protocol introduces a pulsed-current–constant-current (PCCC) strategy between 47% and 58% SOC. This fluctuation-based control provides brief relaxation intervals that promote lithium-ion redistribution and thermal equilibration across the electrodes. These effects reduce concentration polarization and suppress localized overpotential buildup, which in turn alleviates heat accumulation and improves charge uniformity across the cell. After this PCCC stage, the HI5 transitions to a sequence of short constant-current phases and eventually to a gradual current ramp-down as the charge approaches completion.

These examples underscore the complexity and flexibility required in designing fast-charging protocols for batteries. Achieving optimal performance necessitates a thorough understanding of the intricate reaction mechanisms that take place during the charging process. This includes detailed insights into material properties, electrochemical reactions, lithium-ion diffusion kinetics, and the influence of heterogeneity, polarization, and thermal effects on overall battery behavior. To further explore this topic, in the next section, we delve into the fundamental mechanistic factors that underpin the development and customization of efficient and reliable battery charging protocols.

3 | Mechanistic Considerations for Battery Charging Protocol Design

To optimize lithium-ion battery charging protocols, it is essential to thoroughly understand the journey of lithium ions from the cathode, through various interfaces, to their integration into the anode. The effectiveness of this process is influenced by multiple factors, such as the intrinsic material properties, the kinetics of electrochemical reactions, and the heat generation and accumulation within the cell [42, 43]. However, a considerable gap remains between current efforts in battery charging protocol design and a thorough understanding of these fundamental processes, which limits the optimization of fast charging strategies. Here we discuss how these critical factors impact the design of the charging protocol and identify the necessary characterization methods for a comprehensive investigation of each.

3.1 | Battery Materials' Physical and Chemical Properties Change Upon Charging

As the battery's SOC changes, the physical and chemical properties of the battery active materials evolve. For example, the lattice of the active material may expand or contract, and its bulk and interfacial ionic and electronic conductivities vary accordingly [44–49]. The electrode's lithium storage capacity also changes with SOC. Collectively, these changes can impact the efficiency

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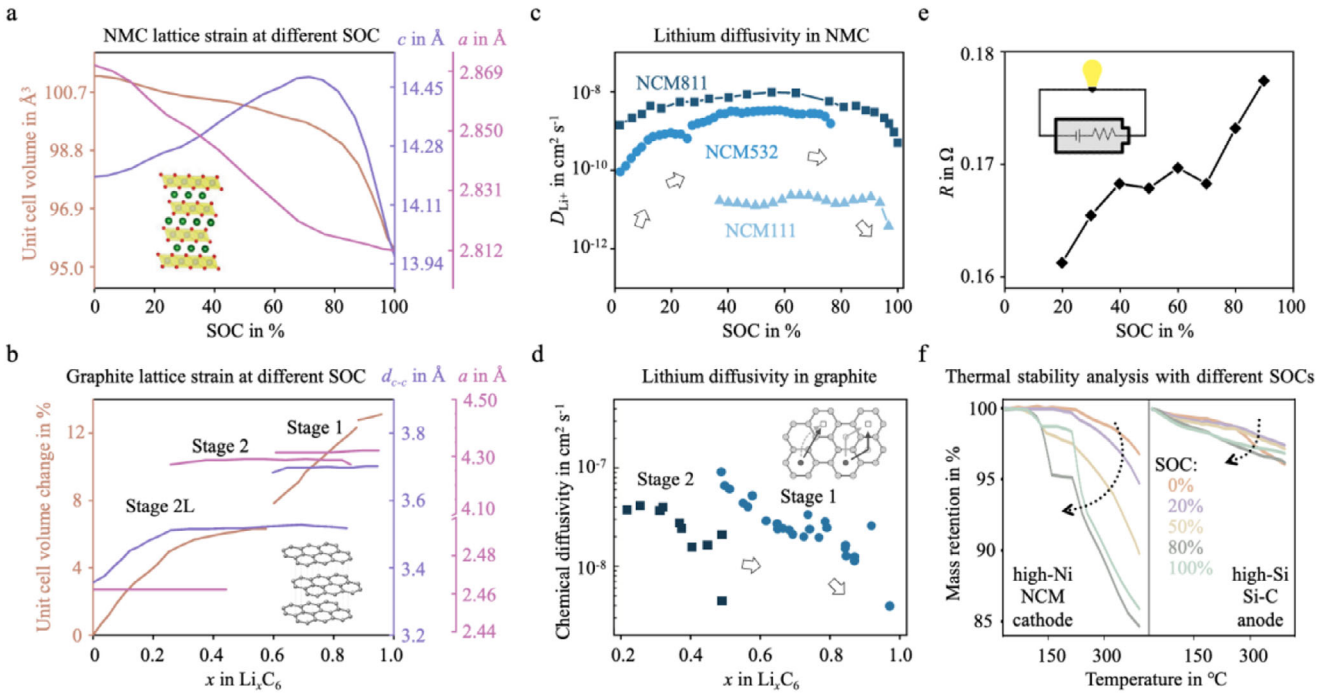


FIGURE 3 | a) Changes in the unit cell volume, the *c* and *a* lattice parameters of NMC811 at various SOCs during the first charge. Reproduced with permission [45]. Copyright 2019, ACS Publications. b) Changes in the unit cell volume, the interlayer spacing $d_{c\text{-}c}$ and the *a*-axis of graphite versus the lithium content. Reproduced with permission [46]. Copyright 2018, ACS Publications. c) Variation of the lithium diffusion coefficient, D_{Li^+}, of NMC811, NMC532, and NMC111 as a function of the SOC. Reproduced with permission [47]. Copyright 2020, RSC. Reproduced with permission [50]. Copyright 2012, Elsevier. Reproduced with permission [51]. Copyright 2013, Elsevier. d) The lithium diffusivity for the Stage 2 (dark blue squares) and Stage 1 (light blue circles) phases in the Li-graphite system, derived from kinetic Monte Carlo simulations. Reproduced with permission [52]. Copyright 1996, Elsevier. e) Battery internal Ohmic resistance at various SOC. Reproduced with permission [53]. Copyright 2015, Wiley. f) Thermal stabilities of Ni-rich NMC cathode and Si-rich Si-C anode at different SOCs. Reproduced with permission [54]. Copyright 2020, Wiley.

of the charging process and need to be understood and integrated into the charging protocol.

3.1.1 | Cathode Evolution During Charging

We first focus on the cathode materials. One of the most obvious charging-induced evolutions of the LIBs material properties is the active cathode's lattice deformation. Layered cathode materials, e.g., $LiNi_xMn_yCo_zO_2$ (NMC), are among the most broadly used cathode materials, featuring a hexagonal structure with (R-3m) symmetry [55, 56]. During charging, lithium ions are extracted, increasing the repulsive force between oxygen layers and expanding the c-axis parameter. Meanwhile the a and b-axis parameters decrease due to M—O bond contraction. Figure 3a shows the evolution of lattice strain (c-lattice, a-lattice, and the unit cell volume) of NMC811 as a function of SOC. Upon charging, the c-axis length increases progressively until the SOC gets to ~70%, reaching a maximum of approximately 14.45 Å. Beyond this point, the c-axis length is negatively correlated with the SOC, and it decreases rapidly as the battery approaches the fully charged state. Conversely, the a-axis length decreases monotonically upon charging, but its decrease slows down at around 70% SOC. As a result, there is a monotonic decrease in NMC's unit cell volume upon charging. This decrease is gradual initially but becomes more pronounced at around 75% SOC and higher. This non-linear, anisotropic lattice strain leads to diminished chemical and structural stability at high SOC. It is closely associated with several detrimental side reactions including oxygen release, dissolution of transition metal ions, and the fracture of active cathode particles [57–60]. The structural instability at high SOC prevents the materials from effectively withstanding the large current and the associated mechanical stress. This is consistent across several commercialized LIBs cathode materials and is one of the key reasons why all the practical fast-charging protocols reduce the current as SOC increases. Additionally, the lithium ion's mobility within the cathode lattice can further constrain the high-rate charging capability [61]. The lithium-ion diffusivity in active cathode materials varies as a function of SOC (see Figure 3c). At very low SOC, the lithium ion's diffusivity is low. This is because the diffusion of lithium ions in the layered lattice structure relies on the exitance of lithium vacancies. A lithium ion can hop into a vacancy site, leaving a vacancy behind, which will accommodate the next lithium ion's movement. At a very low SOC, the population of lithium vacancy is low and, therefore, effective lithium-ion diffusion is hindered. As SOC rises, the concentration of lithium vacancies increases, and vacancy deficiency rapidly ceases to be a limiting factor. As the SOC passes the mid-range and approaches the fully charged state, the lithium-ion diffusivity quickly decreases. This is associated with the lattice unit cell contraction, which increases the energy penalty for lithium ion hopping. The low lithium-ion diffusivity at both the low and high ends of the SOC range significantly hinders the cell's ability to handle high charging currents. As a result, there is usually a current ramp-up phase at very low SOC, while, at high SOC, the current needs to be quickly decreased.

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3.1.2 | Anode Evolution During Charging

On the anode side, similar situations exist. Solvated lithium ions desolvate from the electrolyte, traverse the solid electrolyte interphase (SEI) layer, intercalate into the graphite layers, and continue to diffuse within the layered graphite structure [62]. Under high-rate charging, lithium-ion transport across the SEI is significantly hindered by increased interfacial impedance, thickening of the SEI, and heightened kinetic barriers associated with the desolvation process. These factors collectively impede efficient lithium-ion transport through the interface, reduce fast-charging efficiency, elevate the risk of lithium plating, and accelerate battery performance degradation [36]. As denoted in Figure 3b, graphite lithiation involves several stages. Stage 1 represents a solid-solution-type lithium intercalation throughout the graphite. In comparison, Stage 2 corresponds to a partially lithiated state, with lithium atoms intercalated between every second layer of carbon atoms. Stage 2L refers to a structure like Stage 2, with the distinction that the lithium ions within the layers are not perfectly ordered. During the lithiation process, the graphite volume gradually expands with increasing SOC, primarily due to the increase in interlayer spacing. As lithium ions intercalate between graphite layers, the stacking structure shifts from AB to AA, resulting in approximately 10% volume expansion and leading to structural instability [46]. This is particularly the case at high SOC, where the risk of microcrack formation increases, compromising the mechanical integrity and electrochemical stability of the electrode [63–65]. This makes graphite less suitable for fast charging at high SOC, as rapid charging exacerbates these instabilities, accelerating battery performance degradation. Besides, as the SOC increases, the amount of inserted Li in graphite increases, leading to stronger repulsive Li—Li interactions, which obstruct the lithium diffusion in graphite particles and further trigger Li plating [66, 67]. Kresse et al. [52]. used ab-initio quantum mechanical calculations, combined with kinetic Monte Carlo simulations, to derive the chemical diffusivity in the lithium-graphite system. As shown in Figure 3d, the lithium-ion diffusivity decreases with the increase in in-plane lithium concentration. These properties of graphite anode materials at high SOC clearly prevent it from accommodating a large current at a highly lithiated state. In addition to the pronounced changes near fully lithiated or delithiated states, the mid-SOC range (≈30%–70%) also exhibits marked structural and transport variations [68, 69]. In this region, staging transitions in graphite (e.g., stage 2–1) and concurrent lattice strain in composite electrodes lead to a sharp drop in lithium diffusivity and heterogeneous phase evolution, which in turn affect subsequent electrochemical kinetics.

As discussed above, the physical and chemical properties of both cathode and anode materials change dynamically with SOC. Notably, lattice strains in both cathode and anode can be non-linear and anisotropic, affecting ionic conductivity and exacerbating harmful side reactions. These effects, combined with the intrinsic conductivity limitations of the battery's internal components—such as electrode materials, electrolyte, current collectors, and separator—contribute to the overall cell's Ohmic resistance (R_i) [70]. Figure 3e illustrates the increasing trend of a cell's Ohmic resistance as SOC rises, showing a pronounced surge at SOC levels of 70% or higher. This phenomenon highlights the inefficiency of high-current charging at high SOC, which can result in excessive heat generation. Furthermore, the thermal stability of both cathode and anode materials deteriorates as SOC increases, as depicted in Figure 3f. These material properties play a pivotal role in the design of efficient and safe battery charging protocols. The trends demonstrated in Figure 3 underpin the charging strategies and limitations shown in Figure 2, offering fundamental insights into why specific charging patterns are implemented.

3.2 | Kinetic Factors under Fast Charging: Polarization and Heterogeneity

In addition to Ohmic resistance, the cell experiences polarization resistance during charging, which stems from both concentration polarization and electrochemical polarization. Concentration polarization occurs due to the difference in lithium-ion concentrations across various interfaces. The spontaneous lithium-ion diffusion from high to low concentration areas could add extra energy barriers to the charging process. Electrochemical polarization arises from reaction kinetics and electrode heterogeneity. The lithium-ion extraction from the cathode and insertion into the anode are both constrained by the intrinsic kinetics and activation barriers of the electrode materials.

Under fast-charging, these kinetic factors become more pronounced, significantly influencing the charging process [71–73]. As lithium ions travel from the cathode to the anode, they must pass through several interfaces: 1) cathode grain boundaries, 2) cathode to cathode/electrolyte interface (CEI), 3) CEI to electrolyte, 4) electrolyte to SEI, 5) SEI to anode, and 6) anode grain boundaries. Each interface presents a concentration gradient, contributing to concentration polarization. The externally applied charging current must overcome these resistances, which are intensified at higher charging rates. Additionally, electrochemical polarization intensifies at higher charging rates, driven by kinetic limitations such as Li^+ desolvation and sluggish ion transport across interphases (e.g., SEI and CEI). These processes hinder lithium-ion transport efficiency. As a result, charging speed and overall performance decrease, and battery lifespan shortens. These polarization effects can be experimentally analyzed using electrochemical cycling data. As an illustrative example, half-cell tests ($LiFePO_4$ (LFP) vs. Li and graphite vs. Li) reveal how kinetic limitations manifest under fast-charging conditions. As shown in Figure 4a,b, the voltage differences between the charging and discharging processes become more pronounced at higher C-rates in both half-cell systems. This hysteresis effect is directly tied to cell polarization, which evolves and increases with current density. For a full-cell configuration, such as an LFP vs. graphite cell, the galvanostatic intermittent titration technique (GITT) is often employed to evaluate lithium-ion diffusivity (Figure 4c), a key factor influencing cell polarization. Specifically, during a pulse charge, the voltage immediately rises by iR (where i is the current and R is the resistance). Subsequently, the voltage continues to increase gradually due to the galvanostatic charge pulse, maintaining a constant concentration gradient. Upon cessation of the charging current, the voltage immediately drops by iR and then slowly decreases as lithium ions redistribute within the bulk material until equilibrium is reached. The data shown in Figure 3c is often derived from GITT experiments. It is evident that the polarization effect can be more significant at the

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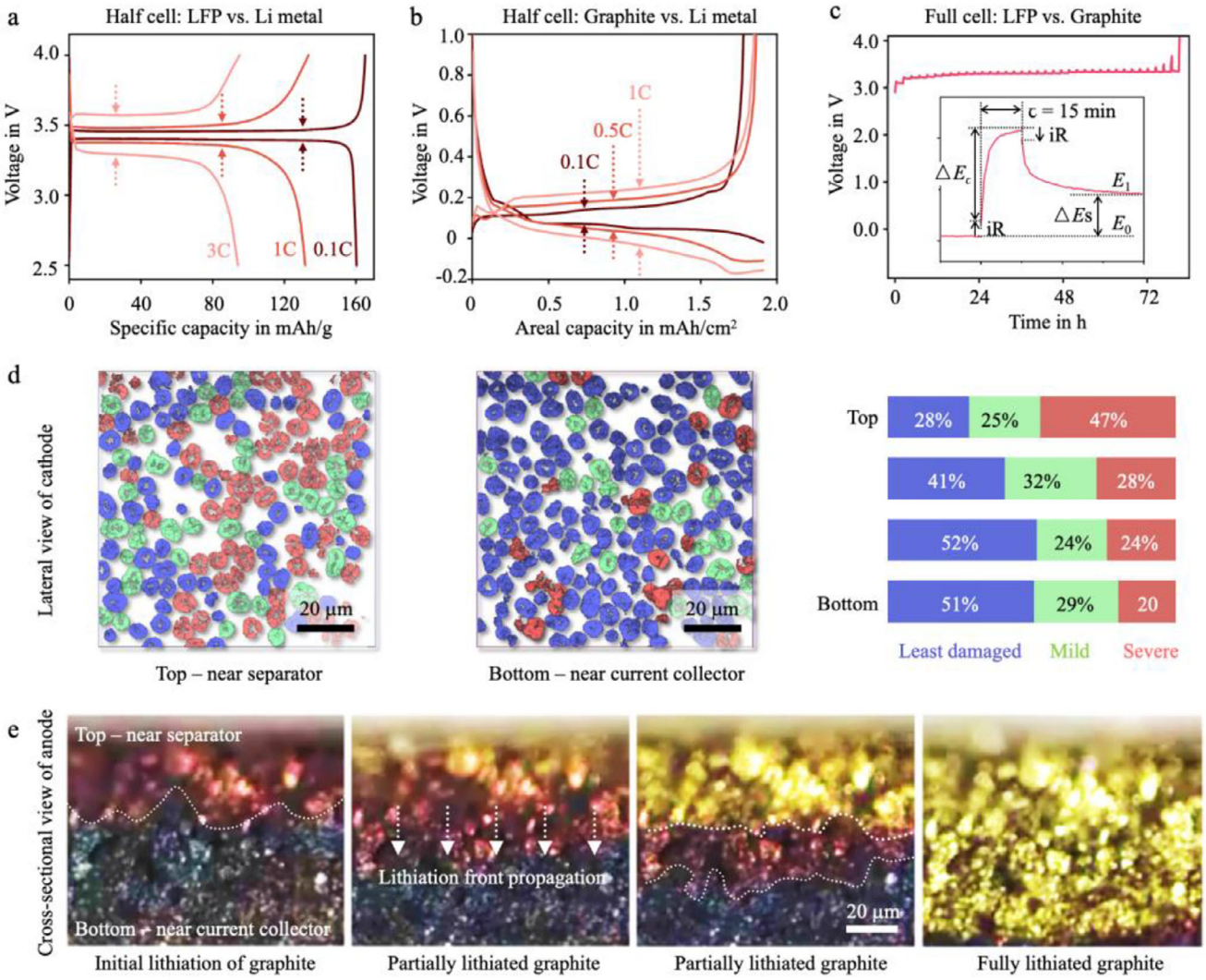


FIGURE 4 | Electrochemical polarization in battery electrodes and cells. a) Half-cell (LFP vs. Li metal) cycling data at various C-rates. b) Half-cell (graphite vs. Li metal) cycling data at various C-rates. c) Full-cell (LFP vs. graphite) GITT data, with an inset illustrating the voltage evolution during a titration step. d) Depth-dependent cathode damage profile revealed by X-ray nano-tomography, inferring the polarization in NMC cathode electrode. Reproduced with permission [78]. Copyright 2019, Wiley. e) Depth-dependent anode SOC evolution revealed by operando optical microscopy, inferring the polarization in graphite anode electrode. Reproduced with permission [79]. Copyright 2023, Elsevier.

very beginning and toward the end of the charging process due to the reduced lithium diffusivity in these SOC regions. However, polarization is not limited to the SOC extremes. Distinct kinetic phenomena also emerge in the mid-SOC range. For example, in graphite anodes, lithium diffusivity decreases sharply near the stage 2–1 transition, while interfacial resistance simultaneously increases due to SEI growth [74]. These coupled effects intensify electrode-level polarization and reaction heterogeneity during fast charging, highlighting the need to account for mid-SOC dynamics in the design of optimized charging protocols.

In addition to these electrochemical tests, the polarization effect can also be directly evaluated using a variety of characterization tools. For example, X-ray tomography has been employed to image the 3D structural disintegration of battery cathode at the particle to electrode levels [75–77]. As shown in Figure 4d, after 10 cycles, the NMC particles near the top and the bottom regions of a well-formed cathode electrode feature significant difference in their damage degrees. This is evident in the 2D lateral virtual slices (left and middle panels of Figure 4d) as well as the statistical analysis shown on the right panel of Figure 4d. Similarly, the depth dependent lithium concentration profiles have been demonstrated in graphite anode electrode using operando optical microscopy. Upon lithiation, graphite changes its color and, therefore, the color can be used to fingerprint the electrode's SOC distribution. In the cross-sectional view of a graphite anode electrode (Figure 4e), the lithiation process initiates at the top of the electrode (near separator) and propagates toward the bottom region (near current collector), demonstrating a complicated and heterogeneous lithiation front as shown in Figure 4e.

The reaction heterogeneity and polarization become more evident during fast charging [80]. This can lead to co-existence of localized insufficient charging and overcharging [81], which in turn triggers undesirable side reactions. When the cell nears full charge, even a brief high-current pulse may cause local overcharging. This pushes the electrode potential beyond its safe operating window. On the cathode side, this could cause

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local phase reconstruction, oxygen release, and structural disintegration. On the anode side, a likely consequence is the lithium plating instead of the desired lithium intercalation into the graphite structure. These irreversible effects are highly detrimental to battery integrity and can severely reduce the cell's overall lifespan. It is useful to highlight that the reaction heterogeneity is a multiscale phenomenon and, once it is formed, it could persistent even after prolonged relaxation of the cell [82]. While electrochemical polarization is typically characterized at the electrode scale, it can also manifest at the single-particle scale, where reaction heterogeneity has been widely reported [83–86]. The degree of microscopic inhomogeneity increases at the charging rate increases [87]. This is dependent on the intrinsic material formations and properties, and it must be taken into consideration for charging protocol design and optimization.

These kinetic factors evolve as a function of the charging rate and SOC. For example, the intrinsic lithium diffusivity in both cathode and anode becomes lower at high SOC, exacerbating the polarization and heterogeneity [51]. To account for these effects, fast-charging protocols need to be designed to minimize reaction heterogeneity by ensuring more uniform current distribution and temperature control across the electrode surface. For example, pulse charging methods offer a promising approach to mitigating charge heterogeneity and reducing polarization within lithium-ion batteries, potentially enhancing both performance and longevity without compromising the charging efficiency [88]. The design of these pulse structures presents numerous degrees of freedom, allowing for a highly adaptable and customizable charging protocol. A key aspect is the resting period that follows each charging pulse. By introducing sufficient rest intervals, charge redistribution within the composite electrode can occur more evenly, preventing localized charge buildup and allowing the system to recover from temporary imbalances. This redistribution can help address spatial heterogeneities within the electrode, ensuring a more uniform state of charge across the material. In addition to the resting periods, another advanced modification to the pulse structure is the incorporation of small, reversed current pulses [89, 90]. These reversed pulses can serve as a tool to counteract kinetic limitations by redistributing ions or alleviating local overpotentials, effectively modulating the cell's electrochemical behavior. This technique can also help in mitigating side reactions, such as lithium plating on the anode [89], which are more likely to occur during high-rate charging. By briefly reversing the current, the cell can experience a "reset" of sorts, helping to stabilize the electrochemical environment and thus promoting a healthier charging process. The design and optimization of pulse charging structures depend heavily on an in-depth understanding of the underlying kinetic mechanisms at play within the battery system [88]. The dynamics of ion diffusion, charge transfer resistance, and electrode degradation are all critical factors that must be considered when tailoring pulse characteristics such as pulse duration, current magnitude, and the length of the rest periods. With detailed insights into these kinetic phenomena, we can potentially fine-tune pulse charging protocols to minimize degradation, enhance energy efficiency, and extend battery lifespan.

In addition to pulse-based strategies, several other approaches have been developed to alleviate electrode polarization. Temperature-managed charging, as demonstrated by integrating rapid internal heating within the cell, effectively decouples charge-transfer kinetics from ambient temperature and enables uniform, lithium-plating-free fast charging across a wide temperature range [16]. Electrolyte and interfacial characteristics, such as solvent composition, salt chemistry, and interphase stability, strongly influence the cell's polarization behavior during fast charging [91]. Therefore, protocol optimization should be tailored to the specific electrolyte–electrode system to ensure efficient ion transport and stable interfacial kinetics. Electrode structural features, including porosity distribution, particle morphology, and conductive network design, also affect lithium-ion diffusion pathways and concentration gradients [92, 93]. Understanding these intrinsic electrode properties enables the rational tuning of charging current profiles to minimize local overpotentials and promote uniform reaction fronts. Furthermore, data-driven and model-predictive control approaches enable real-time adjustment of charging currents based on electrochemical and thermal feedback, providing adaptive responses to transient polarization [94]. Collectively, these complementary considerations, including thermal management, electrolyte–electrode interactions, electrode architecture, and intelligent control, offer an integrated framework for mitigating polarization and improving the safety, stability, and energy efficiency of lithium-ion batteries under fast-charging conditions.

3.3 | Heat Generation and Accumulation

As polarization and SOC-dependent transport limitations intensify at high charging rates, they lead to increased overpotentials and resistive losses within the electrodes. These losses are converted into heat, making heat generation and accumulation another critical factor to consider during fast charging [95]. During fast charging, heat is produced as an unwanted byproduct. The faster the charging rate, the more significant the heat generation becomes. During the charging process of lithium-ion batteries, heat is produced from two primary sources: reversible heat (reaction heat) and irreversible heat (Joule heat) [96]. Reversible heat stems from entropy changes during the electrochemical reactions, determined by the current and the entropic potential, though its overall contribution is relatively small. On the other hand, irreversible heat, dominated by Joule heating, is a result of the operating current and internal resistance, becoming more prominent as charging currents increase. As current flows through the battery's internal components, it encounters resistance, which leads to the dissipation of energy as heat. If this heat is not effectively managed and dissipated, it can cause a sharp increase in the battery's internal temperature, leading to a variety of negative effects [97, 98].

Before discussing how heat generation influences fast-charging protocol design, we provide illustrative measurements of battery temperature evolution under several common charging protocols. The measurements were performed on newly purchased commercial 18650-type NMC–graphite lithium-ion cells (nominal capacity 3.0 Ah, nominal voltage 3.6 V, unused) and a commercial smartphone, with temperature monitored at the cell surface using a thermocouple. Figure 5 presents representative results for the three cylindrical cells and the smartphone. First, the battery temperature is closely correlated with the charging current. For

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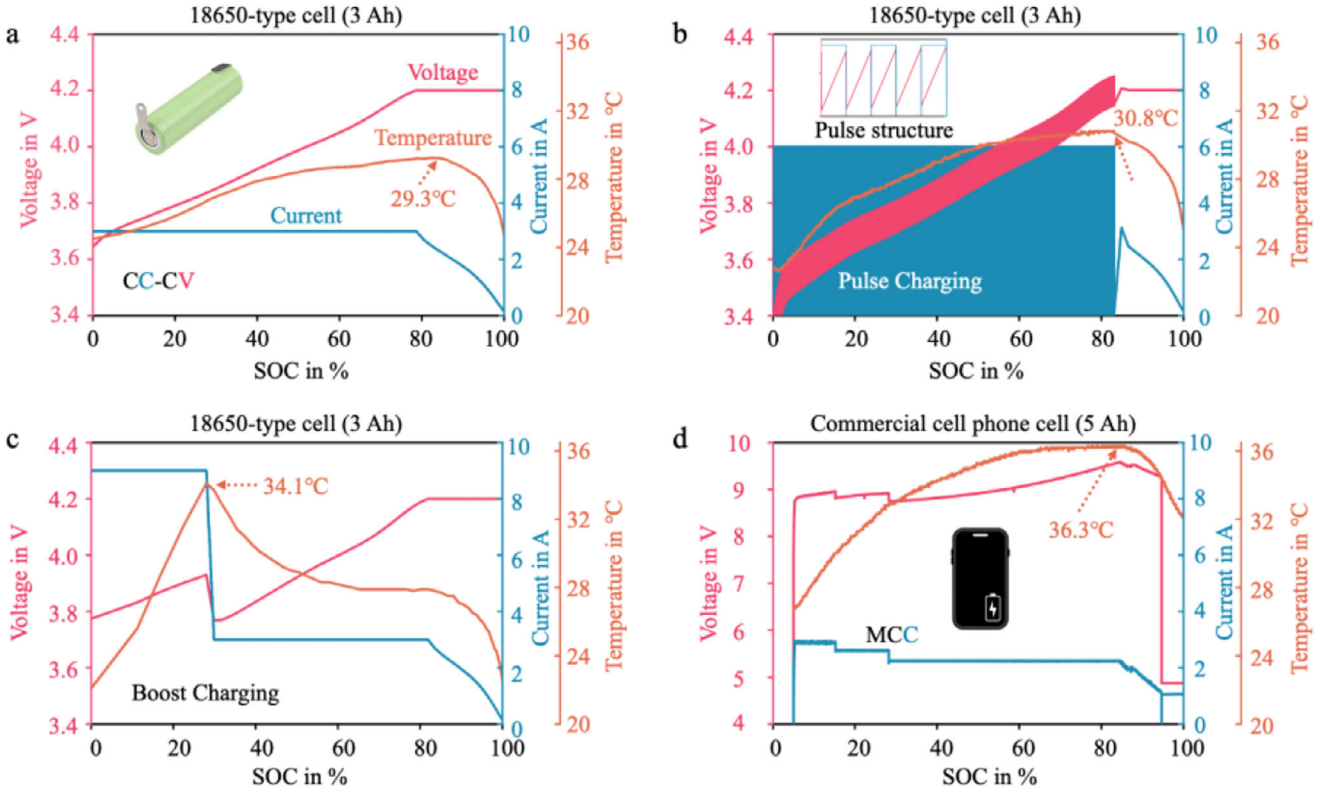


FIGURE 5 | Illustrative measurements of battery temperature evolution under different charging protocols. a) 1C CC–CV for a newly purchased commercial 18650-type NMC–graphite lithium-ion cell (nominal capacity 3.0 Ah, nominal voltage 3.6 V, unused), b) 1C pulse charging (2C current, alternating 1 s charging with 1 s rest) followed by CV for the same type of cell, c) 1C boost charging with CV for the same type of cell, and d) a commercial Xiaomi Redmi Note 12 smartphone (built-in 5.0 Ah cell) charged with a 33 W charger. Temperature was monitored at the cell surface using a thermocouple.

the 18650-type cells charged with CC-CV, pulse charging, and boost charging, the maximum temperature is reached at the end of the periods with the highest current. In both Figure 5a,b, the cell temperature starts to decline when it transitions into the CV phase with current ramp-down. In the boost charging case (Figure 5c), the temperature drops as the cell exits the initial high-rate CC phase. It then stabilizes at a secondary equilibrium before the final CV phase. For the smartphone (Figure 5d), multiple CC phases are executed before a current ramp-down process. The temperature continues to rise until the final current ramp-down phase. These complicated temperature profiles are clearly modulated by the charging protocols. Therefore, the heat generation and accumulation need to be carefully considered when designing the charging protocols.

Several of the factors discussed in the previous sections can affect the heat generation. For example, the lithium diffusivity in battery materials changes as a function of the SOC [51]. At the lower end of the SOC, as we have discussed, the lithium-ion diffusivity is low due to the deficiency of lithium vacancy. On the other hand, the cell's Ohmic resistance (Figure 2e) features a rapid increase as the SOC approaches 70% and higher. Therefore, at the high and low ends of the SOC, the Ohmic heating can be quite significant. In addition to the low- and high-SOC regions, the mid-SOC range (≈ 30%–70%) also contributes significantly to polarization and heat generation. Nonlinear variations in lithium transport and phase evolution within electrode materials give rise to heterogeneous reaction kinetics and localized heat accumulation, even at moderate current levels [74]. Accounting for these effects is essential for achieving thermally balanced fast-charging operation. At the beginning of the charging protocol, a current ramp-up phase could mitigate the unnecessary heat generation. At high SOC, large currents should be avoided since considerable amount of heat has already accumulated, causing the entire cell to reach an elevated temperature. Polarization heating is another notable player. Both concentration and electrochemical polarization contribute to heat generation [99]. As polarization increases with faster charging rates, a greater portion of the input energy is dissipated as heat to overcome kinetic and transport barriers. The resulting temperature rise can, in turn, alter interfacial properties and reaction kinetics, potentially exacerbating polarization under certain conditions. This positive feedback loop can lead to significant temperature elevation during fast charging, posing risks to battery performance and safety. Nevertheless, a recent study in Nature demonstrated that transient temperature elevation can, when precisely controlled, substantially enhance ionic conductivity, Li^+ diffusion, and intercalation kinetics, thereby facilitating fast and uniform charging [100]. However, excessive or prolonged high-temperature exposure accelerates parasitic reactions such as SEI growth and electrolyte decomposition, increasing the likelihood of thermal runaway [101–103]. Importantly, that Nature study [100] further showed that careful thermal management, limiting high-temperature exposure to short durations during preheating and fast-charging, combined with a

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thermally stable dual-salt electrolyte ($LiPF_6$ + LiFSI), effectively mitigates electrolyte decomposition and SEI thickening under transient high-temperature operation, thereby maintaining stable and reversible performance. This detrimental effect becomes particularly pronounced at high SOC, where the reduced thermal stability of electrode materials further amplifies the impact of elevated temperature [104, 105]. Since the optimal operating temperature range depends strongly on electrode chemistry, electrolyte formulation, separators, and cooling efficiency, no universal "safe temperature window" can be defined. In practice, maintaining operation below approximately 45°C provides a reasonable compromise between enhanced Li^+ transport and thermal safety [106]. Therefore, temperature-assisted charging must strike a balance between kinetic enhancement and safety by transiently raising the temperature to improve Li^+ transport, followed by rapid cooling to prevent material degradation. By contrast, at low temperatures (typically < 10°C), electrolyte viscosity and interfacial impedance increase substantially, leading to sluggish ion transport and a higher risk of lithium plating [107]. In this case, mild preheating or current modulation strategies can be employed to facilitate charge transfer while maintaining safety.

If heat generation is not adequately managed, these parasitic reactions can intensify, leading to gas evolution, swelling, or even catastrophic failure [108]. To prevent these scenarios, fast charging protocols must incorporate effective strategies for thermal management. This involves real-time monitoring of the battery's temperature and implementing control mechanisms that can adjust the charging rate to ensure the battery remains within safe thermal operating limits. Techniques such as air/liquid cooling [109], or advanced thermal management materials [110] can be employed to help dissipate heat more efficiently. Additionally, algorithms that can predict heat generation based on the battery's SOC, current, and voltage profiles can provide proactive measures to prevent overheating before it occurs [111].

4 | Advanced Characterization for Battery Charging Protocol Optimization

Despite significant advancements in battery charging protocols, a critical gap persists between the optimization of charging processes and a comprehensive, material-level understanding of LIBs behavior. The inherent structural, chemical, and dynamic complexities of LIBs demand detailed characterization at both the material and cell levels to reveal the fundamental reaction mechanisms governing performance. A deeper understanding of these processes is essential for the design of effective fast-charging protocols that improve efficiency, lifespan, and safety.

Although advanced characterization techniques are regularly utilized in battery research, there remains a pressing need for real-time, in situ monitoring during fast charging [112]. This capability is critical for studying batteries under conditions that closely replicate real-world fast charging applications, providing insights into key factors such as temperature gradients, electrode degradation, and phase transitions [113–116]. However, despite substantial advances, current characterization techniques still face intrinsic limitations in temporal and spatial resolution, chemical sensitivity, and environmental compatibility, particularly under dynamic fast-charging conditions. These challenges hinder the ability to fully capture rapid structural and interfacial evolution within electrodes, and make it difficult to correlate local reaction dynamics, heat generation, and interfacial changes with overall electrochemical performance, thereby limiting mechanistic understanding under realistic operating conditions. Without these detailed observations, optimizing charging protocols for practical use is challenging. In this context, we highlight several essential characterization methods and their relevance to battery charging protocol research. We recognize the significant differences in technical requirements between mechanistic research and practical applications, such as in-vehicle sensing. In this section, we focus on the needs within research environments, while the necessity for compact, cost-effective sensors will be addressed in the following Perspectives section.

4.1 | Structural Characterization

In-situ X-ray diffraction (XRD) is a powerful technique for investigating lattice evolution and phase transformations within battery materials during charging, particularly across different SOC [117–120]. By enabling real-time observation of structural changes in electrode materials as lithium ions are inserted or extracted, in-situ XRD offers invaluable insights into lattice breathing and crystallographic shifts throughout the charge-discharge cycle. This method's material-selective nature allows for the simultaneous measurement of signals from the cathode, anode, and other cell components, facilitating a holistic analysis of the system. In addition, XRD mapping at various SOC levels can also reveal cell-level heterogeneity, such as regions of the electrode experiencing phase transformations at differing rates [87]. With this approach, the crosstalk effect between the cathode and anode electrodes can be evaluated under different charging rates. A notable application of this technique includes its use in identifying heterogeneous lithium plating in fast-charging pouch cells [121]. These observations are essential for understanding the electrochemical processes that impact performance and material stability during operation, making XRD a powerful tool for informing the optimization of charging protocols and battery longevity.

4.2 | Kinetics Characterization

Electrochemical testing methods such as cyclic voltammetry (CV), electrochemical impedance spectroscopy (EIS), and the galvanostatic intermittent titration technique (GITT) are widely employed to probe the kinetic properties and electrochemical behavior of batteries. GITT, in particular, provides key kinetic information by applying small current pulses followed by relaxation periods, during which open-circuit voltage recovery is monitored. This method is instrumental in evaluating lithium-ion diffusion coefficients, polarization phenomena, and identifying the rate-determining steps within the cell. However, these techniques offer integrated, cell-level data without spatial resolution, limiting their ability to capture localized effects. To address this, in-situ transmission X-ray microscopy (TXM) offers a valuable

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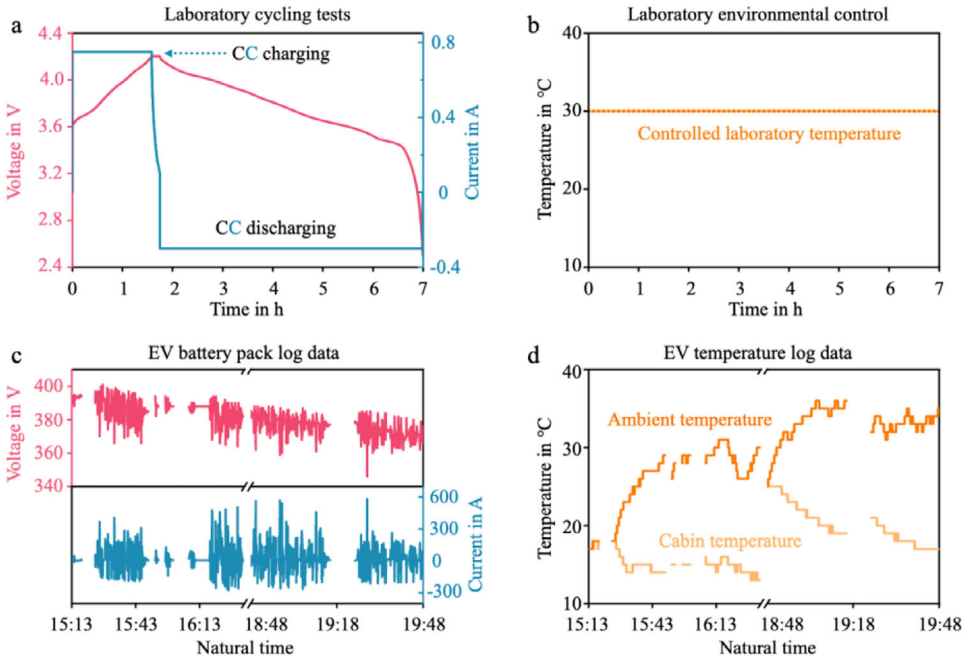


FIGURE 6 | Laboratory controlled battery duty cycle and the duty cycles in electric vehicle in real-world service. a) A standard CC-CV charging and CC discharging duty cycle for a newly purchased commercial 18650-type NMC–graphite lithium-ion cell (nominal capacity 1.5 Ah, nominal voltage 3.6 V, unused). b) The corresponding temperature duty cycle under controlled laboratory environment. c) The voltage and current log data of an EV in real-world service. d) The corresponding temperature log data from this EV.

complement by providing high-resolution imaging at the particle level [122–125]. TXM enables real-time visualization of morphological changes, crack formation, and phase transitions within individual electrode particles during charge and discharge. By combining the electrochemical insights from GITT with the structural data from TXM, we can gain a more comprehensive understanding of the relationship between microstructural evolution and battery performance, facilitating the development of more effective fast-charging strategies. Similarly, as it has been demonstrated, operando imaging with optical microscope can also be very effective in visualizing the kinetic characteristics at the electrode to particle levels [9, 81, 126].

4.3 | Thermal Characterization

Heat generation during fast charging presents a major challenge, as elevated temperatures can accelerate degradation, reduce battery lifespan, and increase the risk of safety hazards such as thermal runway. Standard temperature measurement tools, such as thermocouples and infrared cameras [127–129], are used to monitor surface temperature profiles. While useful, these methods are limited to surface-level measurements and do not capture the internal temperature gradients that can significantly impact battery performance. To address this limitation, techniques such as X-ray diffraction computed tomography (XRD-CT) [130, 131] have emerged as crucial tools for assessing internal thermal behavior. XRD-CT provides high-resolution insights into internal temperature distributions and thermal expansion within the battery, enabling detailed mapping of thermal gradients and their impact on performance. Combined with surface-level thermal imaging, this technique allows for a more complete understanding of heat transfer in LIBs, informing the design of better thermal management strategies to mitigate heat accumulation and enhance battery safety.

The integration of these advanced characterization techniques is essential for bridging the gap between material-level understanding and practical charging protocol optimization. By offering real-time, in-situ insights into the structural, kinetic, and thermal behavior of batteries, these methods enable researchers to refine fast-charging strategies, ultimately improving battery performance, safety, and longevity for applications such as electric vehicles.

5 | Perspectives

The design of fast-charging protocols for lithium-ion batteries hinges on the integration of multiscale material properties. At the microscopic level, changes in crystal structure, ion diffusion, and phase transitions in electrode materials during charging significantly affect battery stability and performance. For instance, lattice contraction in cathode materials at high SOCs leads to mechanical stress, increasing the risk of material degradation over time. At the macroscopic scale, efficient thermal management systems are crucial in preventing excessive heat buildup, which can accelerate material degradation and introduce safety risks like thermal runaway. Optimizing these properties across scales may ensure that both micro-level material behavior and macro-level system performance are well-managed for improved fast-charging efficiency and safety.

The current battery tests are often conducted under controlled laboratory conditions (Figures 6a,b) that do not adequately reflect real-world operating environments (Figure 6c,d). Real-world

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scenarios involve fluctuations in temperature, varying loads, and diverse usage cycles, all of which can impact battery performance and safety. Future testing strategies must simulate these real-life conditions to enhance the adaptability and reliability of fast-charging protocols. By designing charging systems that account for such complexities, researchers can develop more robust protocols that ensure optimal performance even under challenging operating conditions. A notable example is a recent study indicating that dynamic cycling can actually extend battery lifespan, challenging conventional wisdom and presenting exciting opportunities for developing innovative fast-charging protocols [132].

An essential aspect of future fast-charging systems is the integration of advanced high-fidelity sensor technologies to enable variable charging protocols (VCP). Next-generation embedded sensors capable of real-time monitoring of critical parameters such as temperature gradients, interfacial impedance, and internal pressure can provide localized and high-resolution feedback for adaptive control. These sensors will allow dynamic adjustment of current or voltage profiles in response to instantaneous cell conditions, thereby minimizing polarization and preventing lithium plating. For example, distributed temperature and impedance sensors can detect uneven heat generation and increasing interfacial resistance, while acoustic or ultrasonic sensors can identify internal gas evolution and electrode degradation before performance loss occurs [133–135]. When integrated into closed-loop VCP frameworks, these sensing networks can supply high-frequency and multi-parameter data for real-time decision-making and predictive control. The incorporation of such advanced sensors will play a pivotal role in extending battery life, enhancing performance, and ensuring safety during fast charging.

Finally, the use of artificial intelligence (AI) and data-driven optimization methods may significantly enhance the development of fast-charging protocols [136]. By analyzing large datasets, machine learning models can identify patterns and optimize charging processes in real-time, improving both the speed and efficiency of charging. However, ensuring the interpretability of these AI models is crucial for their practical implementation, as transparency and trust in the system are necessary for widespread adoption. By combining AI with real-time sensor data and multiscale material insights, future fast-charging systems may strike a balance between speed, safety, and long-term battery health, supporting advancements in electric vehicles and energy storage technologies.

Through innovations across material design, testing methods, sensor integration, and AI-driven optimization, future fast-charging technologies may achieve the desired balance of speed, safety, and longevity. These advances may lay the groundwork for the next generation of energy storage solutions, making rapid charging more efficient and reliable in diverse applications.

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Conflicts of Interest

The authors declare no conflicts of interest.

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